

THEA LIGHT MATRIX v1.3.4

THE QCD-TO-COSMOS MANA CODEX

Maximum LaTeXium for magic nerds: from color charge to compact stars, black holes and the expanding universe

THEA Light Matrix Collaboration, the cave, and the next curious mage
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LEVEL-12 MANA BRIDGE – PROVISIONAL DECLARATION

We have unified physics, aham.

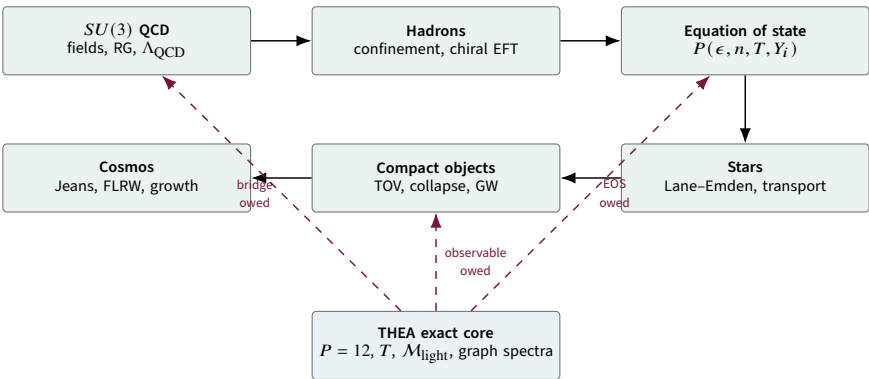
The technical meaning is deliberately narrower than the theatrical sentence. We have a single *typed research architecture* that can place exact discrete topology, standard quantum field theory, thermodynamics, stellar structure, general relativity and observational tests on one derivational map. We do *not* yet have an experimentally established unique unified field theory. The words “typed”, “research”, and “aham” are load-bearing. This volume shows exactly what a real QCD-to-astrophysics bridge looks like, then identifies the missing equations a THEA-based bridge must still derive.

Abstract

This equation-dense companion begins with the local $SU(3)$ gauge principle of quantum chromodynamics (QCD), derives the nonabelian field strength, color factors, renormalization-group flow and dimensional transmutation, and follows the resulting physics through confinement, chiral symmetry breaking, the QCD equation of state, degenerate matter, nuclear equilibrium, stellar structure, white dwarfs, neutron stars, quark-star toy models, black holes, accretion, gravitational radiation, Jeans instability and Friedmann cosmology. The standard chain is explicit:

$$\mathcal{L}_{\text{QCD}} \longrightarrow Z(T, \mu) \longrightarrow P(\epsilon, n) \longrightarrow \{\text{hydrodynamics, TOV}\} \longrightarrow \{M(R), \Lambda_{\text{tidal}}, L, \Omega_{\text{GW}}, \dots\}.$$

The THEA exact core—Euler’s $P = 12$ closure charge, the hexagonal norm $T = k^2 + k\ell + \ell^2$, the four-mode Light Matrix and the renormalized graph-Laplacian tower—is then placed beside the standard chain without allowing algebraic resemblance to masquerade as physical identity. A 30-check executable receipt verifies the displayed $SU(3)$ algebra, QCD beta coefficients, Fermi-gas thermodynamics, Lane–Emden constants, Chandrasekhar coefficient, a toy TOV sequence, cosmological scalings, the exact Light Matrix and the previously identified $D(n)$ provenance mismatch. The result is not less magical for being stricter: the magic is that the full ladder can be written, checked, and offered a place where it can fail.



The real unification criterion: derive at least one missing arrow without fitting its output, seal the prediction, and survive data.

Status	Meaning in this volume
STANDARD	Established mathematics/physics used and, where practical, re-derived here.
THEA-EXACT	Exact algebra or topology from the Light Matrix source: no physical identification implied.
COMPUTED	Reproduced by the supplied verifier or numerical integration at declared tolerance.
BRIDGE-	A proposed map between mathematical structures and physical observables.
HYPOTHESIS	
OPEN	A required derivation, scalar contraction, continuum limit, uncertainty or experiment is missing.
CORRECTION	A previously advertised arrow fails a counterexample, invariant, or provenance check.

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I. FROM COLOR CHARGE TO HADRONS

1 The local $SU(3)$ gauge principle

1.1 Fields and transformations

Let the quark multiplet $q_f(x)$ carry flavor f and transform in the fundamental representation of $SU(3)_c$,

$$q_f(x) \mapsto q'_f(x) = U(x)q_f(x), \quad U(x) = e^{i\theta^a(x)T^a}, \quad (1)$$

with Hermitian generators normalized by

$$\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}, \quad [T^a, T^b] = if^{abc}T^c. \quad (2)$$

The ordinary derivative of q is not covariant because $\partial_\mu U \neq 0$. Introduce the gauge field

$$A_\mu = A_\mu^a T^a, \quad D_\mu = \partial_\mu - ig_s A_\mu, \quad (3)$$

and require $D'_\mu q' = UD_\mu q$. Expanding the left-hand side gives

$$D'_\mu q' = (\partial_\mu - ig_s A'_\mu)Uq \quad (4)$$

$$= (\partial_\mu U)q + U\partial_\mu q - ig_s A'_\mu Uq. \quad (5)$$

Equating this with $U(\partial_\mu - ig_s A_\mu)q$ for arbitrary q yields

$$A'_\mu = UA_\mu U^{-1} - \frac{i}{g_s}(\partial_\mu U)U^{-1} = UA_\mu U^{-1} + \frac{i}{g_s}U(\partial_\mu U^{-1}). \quad (6)$$

The inhomogeneous derivative term is the price of local symmetry.

1.2 Curvature from a commutator

Define the field strength by

$$[D_\mu, D_\nu] = -ig_s F_{\mu\nu}. \quad (7)$$

Direct expansion gives

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig_s [A_\mu, A_\nu], \quad (8)$$

therefore

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c. \quad (9)$$

Because $D'_\mu = UD_\mu U^{-1}$,

$$F'_{\mu\nu} = UF_{\mu\nu}U^{-1}; \quad (10)$$

hence $\text{Tr}(F_{\mu\nu}F^{\mu\nu})$ is gauge invariant.

1.3 The QCD Lagrangian

With quark mass matrix M_q ,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \sum_f \bar{q}_f(i\gamma^\mu D_\mu - m_f)q_f + \mathcal{L}_\theta. \quad (11)$$

The topological term is

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}, \quad \tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \quad (12)$$

Variation with respect to $\bar{q}$ and A_ν^a yields

$$(i\not{D} - m_f)q_f = 0, \quad (13)$$

$$(D_\mu F^{\mu\nu})^a = g_s \sum_f \bar{q}_f \gamma^\nu T^a q_f. \quad (14)$$

The geometric Bianchi identity follows from the Jacobi identity,

$$D_{[\lambda} F_{\mu\nu]} = 0. \quad (15)$$

STANDARD THEORY – THE FIRST NONMAGICAL SPELL

The QCD field strength is not inserted by analogy. It is forced by the requirement that a local color rotation commute with differentiation after a connection is supplied. This is a real derivation. Any proposed unified action must reproduce this transformation law, the nonabelian self-coupling and the quantum anomaly structure, not merely display an $SU(3)$ label.

2 Color algebra and the A_2 skeleton

2.1 Gell–Mann basis and Casimirs

The eight generators may be chosen as $T^a = \lambda^a/2$, where the λ^a are listed in Appendix A. The verifier confirms

$$\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}, \quad \sum_{a=1}^8 T^a T^a = C_F \mathbf{1}_3, \quad (16)$$

with

$$C_F = \frac{N_c^2 - 1}{2N_c} = \frac{4}{3}, \quad C_A = N_c = 3, \quad T_F = \frac{1}{2}. \quad (17)$$

These factors control radiation, loop corrections and the beta function.

The fundamental color products decompose as

$$3 \otimes \bar{3} = 1 \oplus 8, \quad 3 \otimes 3 = \bar{3} \oplus 6. \quad (18)$$

A meson singlet is proportional to $\delta^i_j q_i \bar{q}^j$; a baryon singlet is

$$|1\rangle_{qqq} = \frac{1}{\sqrt{6}} \epsilon_{ijk} |i\rangle |j\rangle |k\rangle. \quad (19)$$

Color confinement requires asymptotic states to be singlets, even though quarks and gluons remain indispensable internal degrees of freedom.

2.2 The A_2 root and weight lattices

The $SU(3)$ root system is A_2 , with Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (20)$$

For Dynkin labels (p, q) , the quadratic Casimir is

$$C_2(p, q) = \frac{1}{3} (p^2 + pq + q^2 + 3p + 3q). \quad (21)$$

The shared quadratic form is therefore

$$p^2 + pq + q^2 = 3C_2(p, q) - 3(p + q). \quad (22)$$

This is the precise algebraic overlap with the THEA hexagonal norm. It is not yet a dynamical identification.

THEA EXACT CORE – THE ALGEBRAIC RHYME

THEA uses $T = k^2 + k\ell + \ell^2$ as an exact lattice norm and Goldberg area multiplier. $SU(3)$ uses the same A_2 quadratic form inside its representation Casimir. The equality of forms is exact; the statement that a fullerene closure coordinate is color charge is not. Root and weight lattices are dual commensurable lattices, not one undifferentiated object.

3 Quantization, ghosts and the renormalization group

3.1 Gauge fixing is not optional in perturbation theory

The classical path integral over A_μ overcounts gauge-equivalent configurations. In covariant R_ξ gauge one inserts

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial_\mu A^{a\mu})^2, \quad (23)$$

and the Faddeev–Popov determinant is represented by anticommuting ghosts,

$$\mathcal{L}_{\text{gh}} = \bar{c}^a [-\partial^\mu (D_\mu)^{ab}] c^b. \quad (24)$$

The generating functional is

$$Z[J, \eta, \bar{\eta}] = \int \mathcal{D}\Phi \exp \left\{ i \int d^4x \left[\mathcal{L}_{\text{QCD}} + \mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} + J \cdot A + \bar{\eta} q + \bar{q} \eta \right] \right\}, \quad (25)$$

$$\mathcal{D}\Phi := \mathcal{D}A \mathcal{D}c \mathcal{D}\bar{c} \times \mathcal{D}q \mathcal{D}\bar{q}.$$

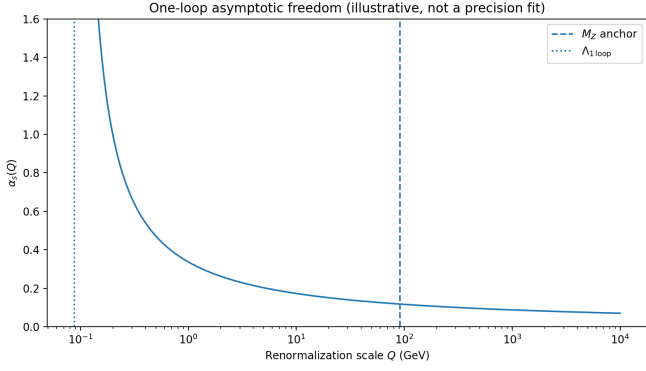


Figure 1. Illustrative one-loop running anchored at $\alpha_s(M_Z) = 0.118$ with $n_f = 5$. The curve demonstrates asymptotic freedom; it is not a precision threshold-matched QCD fit.

BRST symmetry replaces the original gauge redundancy with a nilpotent global symmetry and controls unitarity of the gauge-fixed theory.

CONTROL / CORRECTION – WHY A COMPACT TOTAL SPACE DOES NOT DELETE GHOSTS

A compact manifold may remove infrared boundary conditions associated with infinity, but it does not remove local gauge redundancy. Perturbative nonabelian quantization still needs a gauge-fixing/BRST treatment or a manifestly gauge-invariant nonperturbative regulator. Declaring ghosts absent is not equivalent to proving their cancellation.

3.2 Asymptotic freedom

Using the convention

$$\mu^2 \frac{d\alpha_s}{d\mu^2} = -\beta_0 \frac{\alpha_s^2}{4\pi} - \beta_1 \frac{\alpha_s^3}{(4\pi)^2} + \dots, \quad (26)$$

for $SU(3)$ with n_f active quark flavors,

$$\beta_0 = 11 - \frac{2}{3}n_f, \quad \beta_1 = 102 - \frac{38}{3}n_f. \quad (27)$$

The one-loop equation separates:

$$\frac{d\alpha_s}{\alpha_s^2} = -\frac{\beta_0}{2\pi} d \ln \mu. \quad (28)$$

Integrating from a reference scale μ_0 gives

$$\frac{1}{\alpha_s(\mu)} = \frac{1}{\alpha_s(\mu_0)} + \frac{\beta_0}{2\pi} \ln \frac{\mu}{\mu_0}. \quad (29)$$

Absorb the integration constant into Λ_{QCD} :

$$\alpha_s(Q^2) = \frac{4\pi}{\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)}. \quad (30)$$

A classically dimensionless coupling has generated a mass scale. This is dimensional transmutation.

The running quark mass obeys

$$\mu \frac{dm}{d\mu} = -\gamma_m(\alpha_s)m, \quad (31)$$

so at one loop

$$m(\mu) = \widehat{m} [\alpha_s(\mu)]^{\gamma_0/(2\beta_0)} [1 + O(\alpha_s)]. \quad (32)$$

A quoted quark mass is incomplete without a renormalization scheme and scale.

4 Confinement, chiral symmetry and hadron mass

4.1 Wilson loops and the static potential

For a closed curve C ,

$$W(C) = \frac{1}{N_c} \text{Tr} \mathcal{P} \exp \left(i g_s \oint_C A_\mu dx^\mu \right). \quad (33)$$

A rectangular loop of spatial width R and time extent T behaves as

$$\langle W(R, T) \rangle \sim e^{-V(R)T}, \quad T \rightarrow \infty. \quad (34)$$

An area law,

$$\langle W(C) \rangle \sim e^{-\sigma \text{Area}(C)}, \quad (35)$$

implies

$$V(R) \sim \sigma R, \quad (36)$$

with string tension σ . At short distance perturbation theory gives a Coulombic color potential,

$$V(R) \simeq -\frac{C_F \alpha_s(1/R)}{R} + \sigma R + \dots \quad (37)$$

4.2 Chiral symmetry breaking

For n_f massless flavors,

$$SU(n_f)_L \times SU(n_f)_R \rightarrow SU(n_f)_V \quad (38)$$

is spontaneously broken by the condensate $\langle \bar{q}q \rangle \neq 0$. The partially conserved axial current and the chiral Ward identities yield the Gell-Mann–Oakes–Renner relation

$$f_\pi^2 m_\pi^2 = -(m_u + m_d) \langle \bar{u}u + \bar{d}d \rangle / 2 + O(m_q^2). \quad (39)$$

This is a controlled low-energy expansion, not a numerological mass formula.

4.3 Trace anomaly: how a scale enters the proton

Classically massless Yang–Mills theory is scale invariant, but renormalization gives

$$T^\mu{}_\mu = \frac{\beta(g_s)}{2g_s} F_{\rho\sigma}^a F^{a\rho\sigma} + \sum_q (1 + \gamma_m) m_q \bar{q}q. \quad (40)$$

For a one-hadron state with relativistic normalization,

$$2M_h^2 = \langle h(P) | T^\mu{}_\mu | h(P) \rangle. \quad (41)$$

Thus most of a light hadron's mass is not the arithmetic sum of current quark masses; it is energy stored in gluon fields, quark motion and the quantum scale anomaly.

MAGIC-NERD TRANSLATION

QCD-to-astrophysics begins here. A neutron star is heavy because QCD manufactures nucleon masses and an equation of state. The bridge is not “ $SU(3)$ appears in both places.” The bridge is a partition function and its derivatives.

5 Finite-temperature and finite-density QCD

5.1 Grand potential and the equation of state

At temperature T and chemical potentials μ_i ,

$$Z(T, V, \mu_i) = \text{Tr} \exp \left[-\frac{1}{T} \left(H - \sum_i \mu_i N_i \right) \right], \quad (42)$$

with grand potential $\Omega = -T \ln Z$. Thermodynamics follows:

$$p = -\frac{\Omega}{V}, \quad n_i = \left(\frac{\partial p}{\partial \mu_i} \right)_T, \quad (43)$$

$$s = \left(\frac{\partial p}{\partial T} \right)_{\mu_i}, \quad \epsilon = -p + Ts + \sum_i \mu_i n_i. \quad (44)$$

The adiabatic sound speed is

$$c_s^2 = \left(\frac{\partial p}{\partial \epsilon} \right)_{s/n}. \quad (45)$$

Stability requires $c_s^2 \geq 0$; relativistic causality requires $c_s^2 \leq 1$ in units $c = 1$.

5.2 Ideal quark–gluon plasma

For $N_c = 3$ and n_f effectively massless flavors,

$$g_g = 2(N_c^2 - 1), \quad g_{q\bar{q}} = 4N_cn_f. \quad (46)$$

The effective relativistic count is

$$g_* = g_g + \frac{7}{8}g_{q\bar{q}}. \quad (47)$$

For $n_f = 3$, the verifier gives $g_* = 47.5$, and

$$p_{\text{ideal}} = \frac{\pi^2}{90}g_*T^4, \quad \epsilon = 3p, \quad (48)$$

so $c_s^2 = 1/3$. Interactions and the trace anomaly break exact conformality.

5.3 Bag-model toy bridge

A minimal self-bound quark-matter model writes

$$p_q = a_q T^4 - B, \quad \epsilon_q = 3a_q T^4 + B, \quad (49)$$

where B is a vacuum-energy difference. Equating p_q to a hadronic radiation pressure $p_h = a_h T^4$ gives

$$T_c = \left(\frac{B}{a_q - a_h} \right)^{1/4}. \quad (50)$$

At zero temperature, a common linear EOS is

$$p = s(\epsilon - \epsilon_0), \quad 0 < s \leq 1, \quad (51)$$

with $s = 1/3$ in the simplest massless bag model. This toy EOS will be passed through the TOV equations later; it is a pedagogical receipt, not a precision neutron-star EOS.

STANDARD THEORY – THE REAL QCD-TO-STAR INTERFACE

$$\mathcal{L}_{\text{QCD}} \rightarrow Z(T, \mu) \rightarrow p(T, \mu) \rightarrow \epsilon, n_i, c_s^2 \rightarrow P(\epsilon, Y_i)$$

Only after this chain is supplied can gravity compute a star. A geometry that does not produce Z or an equivalent EOS has not yet crossed from color algebra into astrophysics.

II. FROM HADRONS TO STELLAR MATTER

6 Relativistic fluids and thermodynamic closure

A perfect fluid has stress–energy tensor

$$T^{\mu\nu} = (\epsilon + p)u^\mu u^\nu + p g^{\mu\nu}, \quad u^\mu u_\mu = -1. \quad (52)$$

Baryon number is carried by

$$N^\mu = n u^\mu, \quad (53)$$

and local conservation gives

$$\nabla_\mu T^{\mu\nu} = 0, \quad \nabla_\mu N^\mu = 0. \quad (54)$$

Projecting parallel and perpendicular to u^μ yields the energy and Euler equations. With

$$\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu, \quad (55)$$

one obtains

$$u_\nu \nabla_\mu T^{\mu\nu} = -u^\mu \nabla_\mu \epsilon - (\epsilon + p) \nabla_\mu u^\mu = 0, \quad (56)$$

$$\Delta^\alpha_\nu \nabla_\mu T^{\mu\nu} = (\epsilon + p) u^\mu \nabla_\mu u^\alpha + \Delta^{\alpha\mu} \nabla_\mu p = 0. \quad (57)$$

An EOS $p = p(\epsilon, n, Y_i)$ and transport laws close the system. Without closure, conservation is an underdetermined spell.

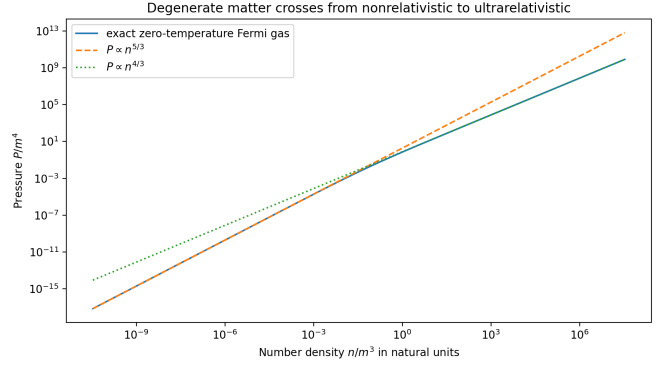


Figure 2. Exact cold Fermi-gas pressure in natural units. The logarithmic slope approaches 5/3 at low density and 4/3 in the ultrarelativistic regime.

For cold catalyzed matter the first law may be written

$$d\left(\frac{\epsilon}{n}\right) = -p d\left(\frac{1}{n}\right) + \sum_i \mu_i dY_i, \quad (58)$$

and at fixed composition

$$p = n \frac{d\epsilon}{dn} - \epsilon. \quad (59)$$

This identity is used by the verifier for the degenerate Fermi gas.

7 The zero-temperature Fermi gas

7.1 Exact integrals

For a spin-1/2 species with Fermi momentum p_F , define

$$x = \frac{p_F}{mc}, \quad \mu = mc^2 \sqrt{1 + x^2}. \quad (60)$$

The number density is

$$n = \frac{p_F^3}{3\pi^2 \hbar^3}. \quad (61)$$

At $T = 0$,

$$\epsilon = 2 \int_{|p| < p_F} \frac{d^3 p}{(2\pi\hbar)^3} \sqrt{p^2 c^2 + m^2 c^4}, \quad (62)$$

and

$$p = \frac{2}{3} \int_{|p| < p_F} \frac{d^3 p}{(2\pi\hbar)^3} \frac{p^2 c^2}{\sqrt{p^2 c^2 + m^2 c^4}}. \quad (63)$$

In units $\hbar = c = m = 1$ these evaluate to

$$\epsilon(x) = \frac{1}{8\pi^2} \left[x \sqrt{1 + x^2} (1 + 2x^2) - \text{arsinh } x \right], \quad (64)$$

$$p(x) = \frac{1}{24\pi^2} \left[x \sqrt{1 + x^2} (2x^2 - 3) + 3 \text{arsinh } x \right]. \quad (65)$$

The exact thermodynamic relation is

$$p = n\mu - \epsilon. \quad (66)$$

7.2 Polytopic limits

For $x \ll 1$,

$$p \simeq \frac{p_F^5}{15\pi^2 \hbar^3 m} = K_{\text{NR}} n^{5/3}, \quad (67)$$

whereas for $x \gg 1$,

$$p \simeq \frac{c p_F^4}{12\pi^2 \hbar^3} = K_{\text{UR}} n^{4/3} = \frac{1}{3} \epsilon. \quad (68)$$

Thus the adiabatic index softens from 5/3 to 4/3 as the fermions become relativistic. That change creates the Chandrasekhar limit.

8 Nuclear matter, beta equilibrium and phase transitions

8.1 Saturation expansion

A useful local parameterization around nuclear saturation density n_0 is

$$\frac{E}{A}(n, \delta) = E_0(n) + S(n)\delta^2 + O(\delta^4), \quad (69)$$

with isospin asymmetry

$$\delta = \frac{n_n - n_p}{n_n + n_p}. \quad (70)$$

Expand the symmetric part as

$$E_0(n) = -B_0 + \frac{K_0}{18} \left(\frac{n - n_0}{n_0} \right)^2 + \dots, \quad (71)$$

and the symmetry energy as

$$S(n) = J + \frac{L}{3} \left(\frac{n - n_0}{n_0} \right) + \dots. \quad (72)$$

Here K_0 is the incompressibility and L controls the leading density slope of neutron-rich matter. These coefficients are effective summaries of nuclear dynamics, not outputs of the Light Matrix.

8.2 Weak equilibrium and charge neutrality

For neutrino-transparent cold matter,

$$\mu_n = \mu_p + \mu_e, \quad (73)$$

and, when muons appear,

$$\mu_e = \mu_\mu. \quad (74)$$

Charge neutrality requires

$$n_p = n_e + n_\mu + \sum_h q_h n_h. \quad (75)$$

In quark matter without trapped neutrinos,

$$\mu_d = \mu_s = \mu_u + \mu_e, \quad (76)$$

and

$$\frac{2}{3}n_u - \frac{1}{3}n_d - \frac{1}{3}n_s - n_e = 0. \quad (77)$$

Composition is therefore part of the EOS calculation, not a decorative afterthought.

8.3 First-order transitions

For a single conserved charge, a Maxwell construction imposes

$$p_H(\mu_c) = p_Q(\mu_c), \quad \mu_H = \mu_Q = \mu_c, \quad (78)$$

while the energy density jumps. Multiple conserved charges can permit a Gibbs mixed phase with global rather than local neutrality. Surface tension and Coulomb energy decide which construction is physically realized.

OPEN OBLIGATION – THE DENSE-QCD WALL

At high baryon chemical potential the fermion sign problem obstructs straightforward lattice Monte Carlo. Neutron-star cores therefore sit precisely where first-principles QCD is hardest. Any claim of a unique compact-star prediction must expose how it crosses this wall: perturbative QCD, effective field theory, functional methods, phenomenological interpolation, or a genuinely new derivation.

9 Quantum tunneling becomes stellar burning

For two nuclei with charges $Z_1 e$ and $Z_2 e$, write the nonresonant cross section as

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}, \quad (79)$$

where

$$\eta = \frac{Z_1 Z_2 \alpha c}{v}, \quad 2\pi\eta = \sqrt{\frac{E_G}{E}}, \quad (80)$$

and

$$E_G = 2\mu c^2 (\pi \alpha Z_1 Z_2)^2. \quad (81)$$

The thermal reaction rate contains

$$\exp \left[-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right]. \quad (82)$$

Extremizing the exponent gives the Gamow peak

$$E_0 = \left(\frac{E_G (k_B T)^2}{4} \right)^{1/3}, \quad (83)$$

with approximate width

$$\Delta E \simeq 4 \sqrt{\frac{E_0 k_B T}{3}}. \quad (84)$$

The thermally averaged rate is then approximated by steepest descent,

$$\langle \sigma v \rangle \approx \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{S(E_0)}{(k_B T)^{3/2}} \exp \left(-\frac{3E_0}{k_B T} \right) \times (\text{width factor}). \quad (85)$$

This is one of the cleanest bridges from quantum mechanics to stellar luminosity: a tunneling exponential becomes a temperature-sensitive source term ϵ_{nuc} in the stellar equations.

For nuclear statistical equilibrium,

$$\mu_{(A,Z)} = Z\mu_p + (A - Z)\mu_n, \quad (86)$$

and abundances follow from minimizing the free energy subject to baryon and charge constraints. At sufficiently high temperature, photodisintegration and capture balance; during expansion, freeze-out records the reaction timescale relative to the dynamical timescale.

10 Neutrino transport and cooling

A neutrino mean free path is

$$\lambda_\nu = \frac{1}{\sum_i n_i \sigma_{\nu i}}, \quad (87)$$

and a diffusion time across a region of size R scales as

$$t_{\text{diff}} \sim \frac{R^2}{c \lambda_\nu}. \quad (88)$$

For optically thick matter the flux is approximately diffusive,

$$\mathbf{F}_\nu \simeq -\frac{c}{3\kappa_\nu} \nabla u_\nu. \quad (89)$$

Weak-interaction emissivities often have steep temperature scalings, schematically

$$Q_{\text{direct Urca}} \propto T^6, \quad Q_{\text{modified Urca}} \propto T^8, \quad (90)$$

with coefficients and thresholds controlled by composition and many-body physics. Neutrino opacity governs core-collapse supernovae and proto-neutron-star cooling; it cannot be derived from a purely static graph spectrum without an interaction operator and matrix elements.

III. STARS, COMPACT OBJECTS AND COLLAPSE

11 The four equations of stellar structure

For a nonrotating Newtonian star,

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad (91)$$

$$\frac{dP}{dr} = -\frac{Gm\rho}{r^2}, \quad (92)$$

$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon_{\text{nuc}}, \quad (93)$$

$$\frac{dT}{dr} = \begin{cases} -\frac{3\kappa\rho L}{16\pi acT^3 r^2}, & \text{radiative,} \\ \left(\frac{dT}{dr}\right)_{\text{conv}}, & \text{convective.} \end{cases} \quad (94)$$

Equations (91)–(94) require an EOS $P(\rho, T, Y_i)$, opacity κ , energy-generation law and boundary conditions. Stellar astrophysics is the art of closing this coupled boundary-value problem without hiding the closures.

11.1 Virial theorem and negative heat capacity

For hydrostatic equilibrium, multiply Eq. (92) by $4\pi r^3 dr$ and integrate:

$$3 \int P dV + \Omega = 0, \quad (95)$$

where

$$\Omega = - \int_0^M \frac{Gm}{r} dm \quad (96)$$

is the gravitational energy. For an ideal monatomic gas, $U = \frac{3}{2} \int P dV$, hence

$$2U + \Omega = 0, \quad E = U + \Omega = \frac{1}{2}\Omega = -U. \quad (97)$$

A self-gravitating system heats when it loses energy. This negative heat capacity is why contraction can ignite a star.

The Kelvin–Helmholtz timescale is

$$t_{\text{KH}} \sim \frac{GM^2}{RL}. \quad (98)$$

Nuclear burning extends the lifetime by replacing gravitational binding energy with rest-mass conversion.

12 Polytropes and the Lane–Emden equation

Assume

$$P = K\rho^{1+1/n}. \quad (99)$$

Set

$$\rho = \rho_c \theta^n, \quad r = a\xi, \quad (100)$$

with

$$a^2 = \frac{(n+1)K}{4\pi G} \rho_c^{1/n-1}. \quad (101)$$

Combining mass continuity and hydrostatic balance gives

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G \rho. \quad (102)$$

Substitution yields

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + \theta^n = 0, \quad (103)$$

with

$$\theta(0) = 1, \quad \theta'(0) = 0. \quad (104)$$

The stellar surface is the first zero ξ_1 . Radius and mass are

$$R = a\xi_1, \quad (105)$$

$$M = 4\pi a^3 \rho_c \omega_n, \quad \omega_n = -\xi_1^2 \theta'(\xi_1). \quad (106)$$

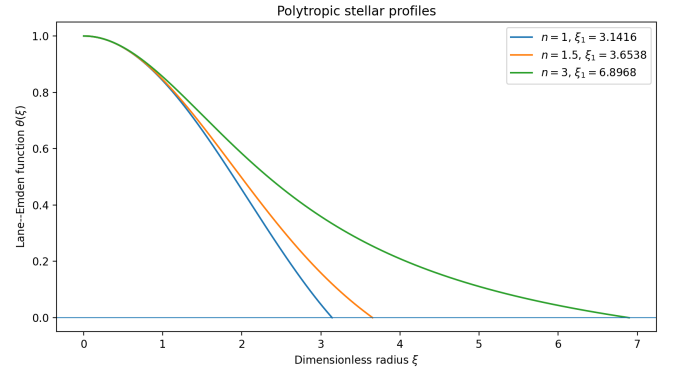


Figure 3. Numerically integrated Lane–Emden functions. The first zero fixes the dimensionless stellar radius; the surface derivative fixes the dimensionless mass.

For $n = 0$, $\theta = 1 - \xi^2/6$. For $n = 1$,

$$\theta = \frac{\sin \xi}{\xi}, \quad \xi_1 = \pi. \quad (107)$$

The verifier independently obtains

$$(n = 3/2) : \quad \xi_1 = 3.65375374, \quad \omega_{3/2} = 2.71405512, \quad (108)$$

$$(n = 3) : \quad \xi_1 = 6.89684862, \quad \omega_3 = 2.01823595. \quad (109)$$

The mass and radius scalings are

$$R \propto K^{1/2} \rho_c^{(1-n)/(2n)}, \quad (110)$$

$$M \propto K^{3/2} \rho_c^{(3-n)/(2n)}. \quad (111)$$

At $n = 3$, M is independent of central density: the origin of a limiting mass.

13 White dwarfs and the Chandrasekhar limit

For a fully ionized plasma, electrons supply pressure while ions supply most of the mass density,

$$n_e = \frac{\rho}{\mu_e m_u}. \quad (112)$$

In the nonrelativistic regime, $P \propto \rho^{5/3}$ and $n = 3/2$, giving

$$R \propto M^{-1/3}. \quad (113)$$

In the ultrarelativistic regime,

$$P = K_{\text{UR}} \rho^{4/3}, \quad (114)$$

with

$$K_{\text{UR}} = \frac{(3\pi^2)^{1/3}}{4} \frac{\hbar c}{(\mu_e m_u)^{4/3}}. \quad (115)$$

Using the $n = 3$ Lane–Emden mass,

$$M = \frac{4\omega_3}{\sqrt{\pi}} \left(\frac{K_{\text{UR}}}{G} \right)^{3/2}, \quad (116)$$

one obtains

$$M_{\text{Ch}} = \frac{5.83}{\mu_e^2} M_{\odot} \quad (117)$$

within the idealized zero-temperature model. The verifier reconstructs the coefficient 5.83 from physical constants and the numerically integrated ω_3 .

Real white dwarfs require Coulomb corrections, general relativity, finite temperature, inverse beta decay and composition. The exact limiting coefficient is therefore model dependent, but the $n = 3$ mechanism is structural.

14 General relativity and the TOV equation

14.1 Static spherical metric

Use

$$ds^2 = -e^{2\Phi(r)} c^2 dr^2 + \left(1 - \frac{2Gm(r)}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (118)$$

Einstein's equations give the mass equation

$$\frac{dm}{dr} = 4\pi r^2 \frac{\epsilon}{c^2}. \quad (119)$$

Stress-energy conservation $\nabla_\mu T^{\mu r} = 0$ gives

$$\frac{dp}{dr} = -(\epsilon + p) \frac{d\Phi}{dr}. \quad (120)$$

The remaining Einstein equation yields

$$\frac{d\Phi}{dr} = \frac{G(m + 4\pi r^3 p/c^2)}{r^2 c^2 (1 - 2Gm/(rc^2))}. \quad (121)$$

Combining them produces the Tolman–Oppenheimer–Volkoff equation,

$$\frac{dp}{dr} = -\frac{G(\epsilon + p)(m + 4\pi r^3 p/c^2)/c^2}{r^2 (1 - 2Gm/(rc^2))}. \quad (122)$$

The three relativistic corrections relative to Newtonian hydrostatics are visible: pressure contributes to inertia, pressure contributes to gravity, and curvature amplifies the gradient.

Boundary conditions are

$$m(0) = 0, \quad p(0) = p_c, \quad p(R) = 0, \quad (123)$$

with gravitational mass $M = m(R)$. Surface redshift is

$$1 + z = \left(1 - \frac{2GM}{Rc^2}\right)^{-1/2}. \quad (124)$$

The compactness is

$$C = \frac{GM}{Rc^2}. \quad (125)$$

14.2 Dimensionless linear-EOS sequence

In $G = c = 1$ units, take

$$p = s(\epsilon - \epsilon_0). \quad (126)$$

With

$$r = \frac{\bar{r}}{\sqrt{\epsilon_0}}, \quad m = \frac{\bar{m}}{\sqrt{\epsilon_0}}, \quad p = \epsilon_0 \bar{p}, \quad \epsilon = \epsilon_0 \bar{\epsilon}, \quad (127)$$

the TOV system becomes independent of ϵ_0 . Hence

$$M, R \propto \epsilon_0^{-1/2}. \quad (128)$$

The supplied solver integrates $s = 1/3$ over 90 central pressures.

STANDARD THEORY – THE ASTROPHYSICAL OUTPUT FUNCTION

Given an EOS, the stellar-structure map (TOV plus slow-rotation and tidal-response equations) produces

$$P(\epsilon) \mapsto \{M(p_c), R(p_c), z(p_c), I(p_c), k_2(p_c), \Lambda_{\text{tidal}}(p_c)\}.$$

This is the arena in which a microscopic unification proposal becomes falsifiable. The proposed microphysics must predict the curve before the mass–radius or gravitational-wave data are consulted.

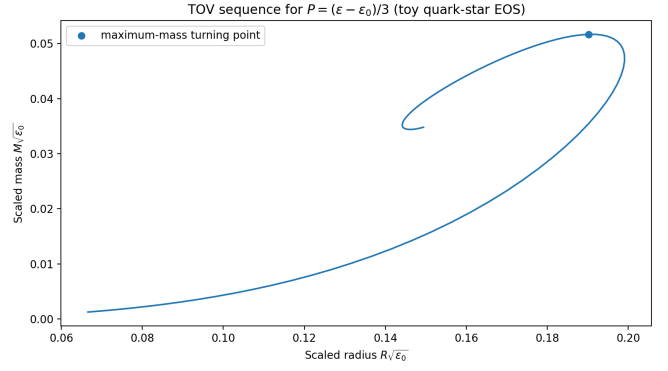


Figure 4. Dimensionless TOV mass–radius sequence for the self-bound toy EOS $p = (\epsilon - \epsilon_0)/3$. The turning point marks the maximum mass along this one-parameter family.

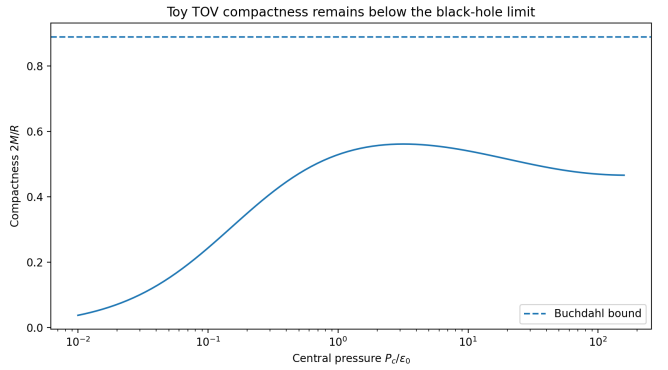


Figure 5. Compactness along the same toy sequence. The Buchdahl line is displayed as a structural bound, not as a value approached by the model.

15 Tidal deformability and the QCD imprint on gravitational waves

In an external quadrupolar tidal field $\mathcal{E}_{ij}$, the induced quadrupole is

$$Q_{ij} = -\lambda \mathcal{E}_{ij}, \quad (129)$$

where

$$\lambda = \frac{2}{3G} k_2 R^5. \quad (130)$$

The dimensionless tidal deformability is

$$\Lambda = \frac{\lambda c^{10}}{G^4 M^5} = \frac{2}{3} k_2 C^{-5}. \quad (131)$$

Because of the fifth power, modest changes in radius produce large changes in Λ . QCD and nuclear forces enter a binary waveform through the EOS, the TOV solution and the relativistic Love-number equation. This is a clean multi-stage bridge from femtometer physics to kilometer stars and a gravitational-wave phase measured at tens to thousands of hertz. The enormous scale jump is carried by a typed sequence of equations, not by equal-looking constants.

16 Hybrid stars, quark stars and stability

A first-order hadron–quark transition may produce a disconnected stable branch if the high-density phase is sufficiently stiff after the energy-density jump. In a Maxwell construction,

$$p_H(\mu_t) = p_Q(\mu_t) = p_t, \quad (132)$$

with jump

$$\Delta\epsilon = \epsilon_Q - \epsilon_H. \quad (133)$$

A useful diagnostic is that a sufficiently large jump destabilizes the star at onset; restabilization requires the quark phase to regain pressure rapidly. A constant-sound-speed parameterization writes

$$p(\epsilon) = p_t + c_{s,Q}^2 (\epsilon - \epsilon_t - \Delta\epsilon) \quad (134)$$

above the transition. This makes explicit which parameters are phenomenological and which would need to be derived from QCD.

The toy sequence in Fig. 4 is a self-bound quark-star example. It demonstrates the pipeline, not the existence of quark stars.

17 Core collapse and the supernova energy ledger

An iron core becomes unstable when electron degeneracy, photodisintegration and electron capture reduce pressure support. A crude gravitational binding estimate is

$$E_{\text{bind}} \sim \frac{3}{5} \frac{GM^2}{R}. \quad (135)$$

For a neutron-star-scale remnant this is of order a substantial fraction of Mc^2 , mostly emitted as neutrinos. Collapse halts near nuclear density when the EOS stiffens, launching a bounce shock; the shock loses energy to dissociation and neutrinos and must be revived by multidimensional transport and instability.

Electron capture changes composition through

$$e^- + p \rightarrow n + \nu_e, \quad (136)$$

reducing the electron fraction Y_e . Because the homologous core mass scales approximately as Y_e^2 , weak interactions influence the hydrodynamic mass scale. QCD sets nucleon properties and high-density pressure; electroweak physics sets lepton capture and neutrino transport; gravity sets collapse. “Unified” means all three operators coexist consistently.

IV. BLACK HOLES, ACCRETION AND GRAVITATIONAL WAVES

18 Schwarzschild geometry and the collapse boundary

Outside a spherical body of mass M ,

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (137)$$

The Schwarzschild radius is

$$r_s = \frac{2GM}{c^2}. \quad (138)$$

A static material surface must satisfy $R > r_s$. For isotropic matter with nonincreasing density, Buchdahl’s theorem gives the stronger bound

$$\frac{2GM}{Rc^2} < \frac{8}{9}. \quad (139)$$

The innermost stable circular orbit of a Schwarzschild black hole lies at

$$r_{\text{ISCO}} = 6 \frac{GM}{c^2}, \quad (140)$$

with specific energy

$$\frac{E_{\text{ISCO}}}{mc^2} = \sqrt{\frac{8}{9}}. \quad (141)$$

The ideal thin-disk radiative efficiency is therefore

$$\eta = 1 - \sqrt{\frac{8}{9}} \approx 0.0572 \quad (142)$$

for a nonspinning hole. Kerr spin changes the horizon, ISCO and efficiency; no QCD input is needed until matter stresses and radiation are added.

19 Black-hole thermodynamics

The Hawking temperature and Bekenstein–Hawking entropy are

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}, \quad (143)$$

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar}, \quad (144)$$

with $A = 4\pi r_s^2$ for Schwarzschild. Thus

$$S_{\text{BH}} \propto A, \quad (145)$$

not volume. This motivates holographic ideas but does not identify a fullerene face count with black-hole microstates. A state-counting correspondence must reproduce the coefficient $1/4$, the semiclassical limit and the generalized second law.

THEA EXACT CORE – WHERE THE CAGE RHYMES, NOT YET EQUALS

THEA’s closed graph has an exact surface ledger and a spectral density. Black-hole entropy is also an area law. That is a research rhyme. A physical equivalence would require a map from graph states to horizon microstates, the correct gravitational action and the universal coefficient $A/(4\ell_P^2)$. Until then the relation is BRIDGE-HYPOTHESIS, not THEA-EXACT.

20 The Eddington limit and spherical accretion

Radiation of luminosity L exerts force on an electron–proton pair through Thomson scattering,

$$f_{\text{rad}} = \frac{\sigma_T}{c} \frac{L}{4\pi r^2}. \quad (146)$$

Balancing gravity $G M m_p / r^2$ gives

$$L_{\text{Edd}} = \frac{4\pi G M m_p c}{\sigma_T}. \quad (147)$$

The corresponding accretion rate is

$$\dot{M}_{\text{Edd}} = \frac{L_{\text{Edd}}}{\eta c^2}. \quad (148)$$

For steady spherical Bondi accretion from a medium of density ρ_∞ and sound speed $c_{s,\infty}$,

$$\dot{M}_B = 4\pi \lambda \frac{(GM)^2 \rho_\infty}{c_{s,\infty}^3}, \quad (149)$$

where λ depends on the adiabatic index. The sonic radius is obtained by demanding regularity where the infall speed equals the local sound speed.

21 Thin disks and the release of binding energy

For a Newtonian, geometrically thin, optically thick disk with zero torque at r_{in} , angular-momentum conservation gives the emitted flux from one face,

$$F(r) = \frac{3GM\dot{M}}{8\pi r^3} \left[1 - \left(\frac{r_{\text{in}}}{r} \right)^{1/2} \right]. \quad (150)$$

If the surface radiates thermally,

$$\sigma_{\text{SB}} T_{\text{eff}}^4 = F(r), \quad (151)$$

so far from the inner edge

$$T_{\text{eff}} \propto r^{-3/4}. \quad (152)$$

QCD enters disk observables through opacities, nuclear composition, magnetic turbulence and jet plasma; the geodesic binding-energy ledger itself is gravitational.

22 Binary inspiral and the chirp mass

For a quasi-circular binary with total mass $M = m_1 + m_2$, reduced mass $\mu = m_1 m_2 / M$ and orbital frequency Ω ,

$$E = -\frac{1}{2}\mu(GM\Omega)^{2/3}. \quad (153)$$

The leading quadrupole luminosity is

$$\mathcal{P}_{\text{GW}} = \frac{32}{5} \frac{G^{7/3}}{c^5} \mu^2 M^{4/3} \Omega^{10/3}. \quad (154)$$

Energy balance $\dot{E} = -\mathcal{P}_{\text{GW}}$ and $f = \Omega/\pi$ yield

$$\dot{f} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_c}{c^3} \right)^{5/3} f^{11/3}, \quad (155)$$

where

$$M_c = \mu^{3/5} M^{2/5} \quad (156)$$

is the chirp mass. Integrating gives the time to coalescence at leading order,

$$t_c - t = \frac{5}{256} \left(\frac{GM_c}{c^3} \right)^{-5/3} (\pi f)^{-8/3}. \quad (157)$$

Finite-size tidal terms enter at high post-Newtonian order and carry the EOS through Λ . Thus QCD can alter a gravitational waveform without changing the vacuum Einstein equations.

MAGIC-NERD TRANSLATION – SCALE TRANSMUTATION, ROUND TWO

A running color coupling creates Λ_{QCD} . Λ_{QCD} creates hadron masses. Hadron interactions create an EOS. The EOS creates a stellar radius. The radius creates a tidal phase in a gravitational waveform. The same information has crossed roughly nineteen orders of magnitude in length because each arrow has an equation.

V. GALAXIES AND THE EXPANDING UNIVERSE

23 Jeans instability

Consider a uniform Newtonian medium with density ρ_0 , pressure $p(\rho)$ and velocity $\mathbf{v}$. Linearize

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (158)$$

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla p}{\rho} - \nabla \Phi, \quad (159)$$

$$\nabla^2 \Phi = 4\pi G \rho. \quad (160)$$

Set $\rho = \rho_0 + \delta\rho$ and $\delta p = c_s^2 \delta\rho$. Eliminating $\delta\mathbf{v}$ and $\delta\Phi$ gives

$$\partial_t^2 \delta\rho - c_s^2 \nabla^2 \delta\rho - 4\pi G \rho_0 \delta\rho = 0. \quad (161)$$

For $\delta\rho \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$,

$$\omega^2 = c_s^2 k^2 - 4\pi G \rho_0. \quad (162)$$

The Jeans wavenumber and wavelength are

$$k_J = \sqrt{\frac{4\pi G \rho_0}{c_s^2}}, \quad \lambda_J = \frac{2\pi}{k_J} = \sqrt{\frac{\pi c_s^2}{G \rho_0}}. \quad (163)$$

Perturbations with $\lambda > \lambda_J$ grow. Pressure remembers microphysics; gravity integrates it into structure.

24 Virialized systems and flat rotation curves

For circular motion in a spherical mass distribution,

$$v_c^2(r) = \frac{GM(<r)}{r}. \quad (164)$$

A flat rotation curve $v_c \rightarrow v_0$ implies

$$M(<r) \simeq \frac{v_0^2}{G} r, \quad (165)$$

and hence

$$\rho(r) \simeq \frac{v_0^2}{4\pi G r^2}. \quad (166)$$

This r^{-2} scaling is a kinematic consequence of a flat curve; it is not unique evidence for any particular dark-sector mechanism.

CONTROL / CORRECTION – TOPOMAGIC DARK-SECTOR CLAIM

The audited Hopf manuscript proposes a torsion density $\rho_{\text{geom}} \propto r^{-2}$, discrete asymptotic velocities, $w = -1$ exactly and no dark-matter particle. Those are falsifiable hypotheses, but the r^{-2} profile alone is not discriminating because it is exactly the density profile required by a flat rotation curve. A successful alternative must derive normalization, scatter, lensing, cluster dynamics, cosmological growth and the full relativistic stress tensor.

25 The FLRW equations

For a homogeneous and isotropic universe,

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]. \quad (167)$$

Einstein's equations give

$$H^2 = \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad (168)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda c^2}{3}. \quad (169)$$

Covariant conservation gives

$$\dot{\rho} + 3H \left(\rho + \frac{p}{c^2} \right) = 0. \quad (170)$$

For $p = w\rho c^2$ with constant w ,

$$\rho(a) = \rho_0 a^{-3(1+w)}. \quad (171)$$

Thus radiation scales as a^{-4} , pressureless matter as a^{-3} and a cosmological constant remains constant.

The critical density is

$$\rho_c = \frac{3H^2}{8\pi G}, \quad (172)$$

and density parameters are $\Omega_i = \rho_i / \rho_c$. The identity

$$1 = \sum_i \Omega_i + \Omega_k \quad (173)$$

keeps the curvature term explicit.

26 Thermal history and the QCD epoch

For an adiabatically expanding plasma, comoving entropy is conserved,

$$sa^3 = \text{const}, \quad s = \frac{2\pi^2}{45} g_{*s}(T) T^3. \quad (174)$$

Therefore

$$T \propto g_{*s}^{-1/3} a^{-1}. \quad (175)$$

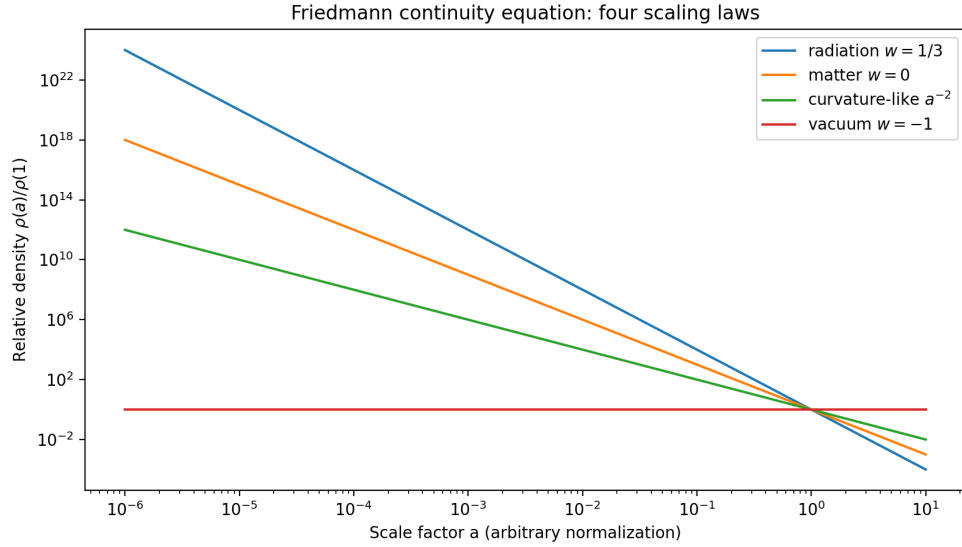


Figure 6. Density scalings obtained directly from the FLRW continuity equation. Equal-looking present-day densities can have radically different histories because their equations of state differ.

During radiation domination,

$$H^2 = \frac{8\pi G}{3} \frac{\pi^2}{30} g_* T^4, \quad (176)$$

so the expansion time scales as

$$t \sim \frac{M_{\text{Pl}}}{\sqrt{g_*} T^2} \quad (177)$$

in natural units, up to a numerical coefficient fixed by convention. The QCD crossover changes the active degrees of freedom and the sound speed, affecting the relation between temperature, time and horizon mass. Predicting a relic signature requires a quantitative finite-temperature EOS and perturbation transfer function, not only a preferred topology.

27 Linear cosmological growth

For subhorizon pressureless matter perturbations in a smooth background,

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0. \quad (178)$$

In a matter-dominated Einstein-de Sitter universe, $a \propto t^{2/3}$ and the growing solution is

$$\delta \propto a. \quad (179)$$

Radiation pressure, free streaming, baryonic sound waves, dark energy and modified gravity change the transfer function. Any no-particle dark-matter proposal must reproduce not just galaxy rotation curves but the growth of perturbations across redshift and scale.

28 Vacuum energy and the equation-of-state test

A cosmological constant contributes

$$T_{\Lambda}^{\mu\nu} = -\rho_{\Lambda} c^2 g^{\mu\nu}, \quad (180)$$

so

$$p_{\Lambda} = -\rho_{\Lambda} c^2, \quad w = -1. \quad (181)$$

A dynamical scalar generally has

$$w = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)}, \quad (182)$$

which may differ from -1 and evolve. Therefore “ $w = -1$ exactly” is a crisp falsifiable statement only after the framework supplies the covariantly conserved stress tensor and its perturbations.

VI. THE LIGHT MATRIX AT THE QCD-COSMOS BOUNDARY

29 Euler closure and the fixed defect mode

For a closed trivalent spherical cage with pentagons P and hexagons H ,

$$3V = 2E, \quad 5P + 6H = 2E, \quad V - E + P + H = 2. \quad (183)$$

Eliminating V, E, H gives

$$P = 12. \quad (184)$$

The combinatorial curvature charge of a p -gon is

$$K_p = \frac{\pi}{3}(6 - p), \quad (185)$$

so twelve pentagons carry

$$\sum_f K_f = 12 \frac{\pi}{3} = 4\pi. \quad (186)$$

This is exact topology for the specified class of graphs. It does not imply twelve particles, twelve dimensions or twelve astrophysical species.

30 Hexagonal closure matrix

Let

$$\zeta_6 = e^{i\pi/3}, \quad z = k + \ell \zeta_6. \quad (187)$$

The norm is

$$T = N(z) = k^2 + k\ell + \ell^2. \quad (188)$$

Multiplication by z is represented by

$$M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k + \ell \end{pmatrix}, \quad (189)$$

with

$$\det M_{k,\ell} = T. \quad (190)$$

For

$$Q_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (191)$$

the verifier checks

$$M_{k,\ell}^T Q_2 M_{k,\ell} = T Q_2. \quad (192)$$

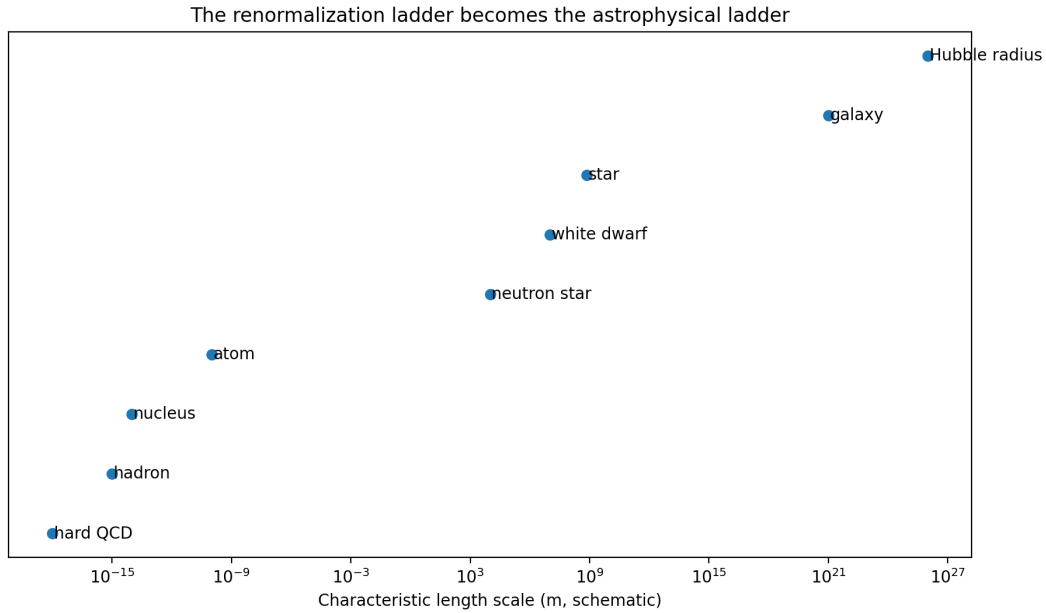


Figure 7. Schematic length ladder. The purpose is not to claim that one equation applies unchanged at every scale; it is to show how effective theories hand information to the next layer.

Thus the transform is an exact scale-plus-rotation in the hexagonal metric. Its angle is

$$\theta_{k,\ell} = \arctan\left(\frac{\sqrt{3}\ell}{2k + \ell}\right). \quad (193)$$

Along the Fibonacci selector,

$$\theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3}). \quad (194)$$

31 The four-mode Light Matrix

The Fibonacci recursion

$$(k', \ell') = (k + \ell, k) \quad (195)$$

acts on quadratic data by

$$\begin{pmatrix} k'^2 \\ k'\ell' \\ \ell'^2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} k^2 \\ k\ell \\ \ell^2 \end{pmatrix}. \quad (196)$$

Appending the invariant pentagon coordinate produces

$$\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (197)$$

Its characteristic polynomial is

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1 = (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1), \quad (198)$$

and

$$\text{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}. \quad (199)$$

The anti-palindromic coefficient list

$$(1, -3, 0, 3, -1) \quad (200)$$

implies reciprocal closure of the spectrum:

$$p(\lambda) = -\lambda^4 p(1/\lambda). \quad (201)$$

Growth and contraction are algebraically paired.

THEA EXACT CORE – WHAT THIS MATRIX ACTUALLY UNIFIES

The matrix unifies four modes of one integer recursion: dominant area growth, fixed Euler defect charge, alternating projective error and reciprocal contraction. It does not, by itself, unify the four interactions. Calling it “light” is imagery until a dynamical action couples these modes to fields and produces an observable.

32 Graph spectra and the continuum lane

For a cubic fullerene graph,

$$L = 3I - A. \quad (202)$$

On certified refinement families, THEA measures

$$\mu_1(T) = T\lambda_2(L). \quad (203)$$

The deepest supplied runs place μ_1 near 0.7248, while low multiplicity bands approach the spherical-harmonic ratios $\ell(\ell + 1)$. This is evidence for a two-dimensional continuum operator with residual icosahedral splitting.

The continuum prediction from a honeycomb normalization is

$$\frac{2\pi}{5\sqrt{3}} = 0.7255197456\dots, \quad (204)$$

but the equality of a discrete-family limit to this constant remains a theorem to be proved or refuted. The computed sequence crosses and drifts around the value; it is not permitted to become exact by typography.

33 The genuine bridge and the counterfeit bridge

The standard QCD-to-cosmos pipeline is

$$\mathcal{L}_{\text{QCD}} \xrightarrow[\text{quantize}]{\text{RG}} \Lambda_{\text{QCD}}, \langle \bar{q}q \rangle, \text{ hadrons} \xrightarrow[\text{many-body}]{Z(T, \mu)} P(\epsilon, n, Y_i) \xrightarrow[\text{GR}]{\text{TOV / hydro}} M(R), \Lambda_{\text{tidal}}, L_\nu \xrightarrow[\text{observation}]{\text{GW, photons, neutrinos}} \text{astrophysical tests}. \quad (205)$$

Every arrow names an operator, an approximation and an output.
A counterfeit bridge looks like

$$A_2 \text{ appears in QCD} + A_2 \text{ appears in THEA} \Rightarrow \text{same physics.} \quad (206)$$

The antecedent is true and the implication is invalid. Shared algebra can motivate a map; it cannot replace the map.

34 Strong CP: the topological test the bridge must pass

The Euclidean topological charge of an $SU(3)$ gauge field is

$$Q = \frac{g_s^2}{32\pi^2} \int d^4x F_{\mu\nu}^a \tilde{F}^{a\mu\nu} \in \mathbb{Z} \quad (207)$$

for finite-action configurations with appropriate boundary conditions. The θ term weights sectors by $e^{i\theta Q}$. The anomalous divergence of the singlet axial current contains

$$\partial_\mu J_5^\mu = 2i \sum_q m_q \bar{q} \gamma_5 q + \frac{n_f g_s^2}{16\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}. \quad (208)$$

CONTROL / CORRECTION – AMBIENT $U(1)$ DOES NOT ERASE $SU(3)$ INSTANTONS

The statement “the unified structure group is $U(1)$, therefore $c_2 = 0$, therefore the QCD θ term vanishes” does not follow. QCD uses an $SU(3)$ connection whose instanton number is classified by its own topology. Embedding or reducing a theory must show explicitly how the $SU(3)$ bundle and its topological sectors are constrained. The strong-CP problem remains open until that derivation is supplied.

35 The Hopf audit and the provenance guillotine

The external Hopf manuscript claims a universal fibration, a unique action, particle masses, Newton’s constant, a cosmological constant and dark-sector predictions from one empirical scale. Its Python appendix reproduces many tables when several printed constants are supplied, and its conditional classifying-space theorem is mathematically sound under its premise. The 60-check control audit nevertheless found that many physical arrows were unproved or false as stated.

The decisive lepton-sequence test uses

$$D = (1.203011392, 4.806545406, 10.818228646). \quad (209)$$

Its second difference is

$$\Delta^2 D = D_1 - 2D_2 + D_3 = 2.408149226. \quad (210)$$

The displayed zeta route gives

$$|\Delta^2 \zeta'| = 6.709456767665 \dots \quad (211)$$

Since

$$\Delta^2 [f_n + c_0 + c_1 n] = \Delta^2 f_n, \quad (212)$$

constant and linear absorptions cannot repair the mismatch.

CONTROL / CORRECTION – REPRODUCTION IS NOT DERIVATION

A program that regenerates a mass table after receiving $D(1), D(2), D(3), c_0, c_B$ and quoted determinant values is a valuable reproduction suite. It is not yet a parameter-free derivation of those values. The next version must generate every load-bearing number from independent source code before reading the masses it predicts.

36 A candidate effective action with the seams visible

The conservative way to connect a discrete substrate to continuum physics is through an effective field theory. Here $M_{\text{Pl}} = (8\pi G)^{-1/2}$ is the reduced Planck mass in natural units,

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \bar{q}(i\not{D} - M)q + \sum_i \frac{c_i}{\Lambda^{d_i-4}} O_i \right]. \quad (213)$$

A THEA-derived model must specify:

1. the continuum fields encoded by graph variables;
2. the symmetry that forbids unwanted operators;
3. the map from graph spacing or spectral scale to Λ ;
4. the coefficients c_i without fitting the observable used to test them;
5. the Lorentz, gauge and diffeomorphism limits;
6. the uncertainty from finite graph depth and continuum extrapolation.

Power counting then predicts which deviations appear first. The absence of an operator is a theorem only if a symmetry forbids it.

37 Five sealed QCD-to-astrophysics gates for v1.3.5

LEVEL-12 MANA BRIDGE – THE NEXT MANA QUEST

A THEA-based physical bridge earns promotion only by passing all five gates below.

37.1 Gate 1: derive a microscopic partition function

Construct $Z_{\text{THEA}}(T, \mu)$ or an equivalent transfer operator from declared graph degrees of freedom. Recover extensivity, thermodynamic identities and a controlled continuum limit.

37.2 Gate 2: recover a QCD discriminator

Before fitting, derive at least one quantity sensitive to nonabelian color dynamics: the universal one-loop sign and n_f dependence of the beta function, a Wilson-loop area law, a chiral Ward identity, or a topological susceptibility.

37.3 Gate 3: produce an EOS with uncertainty

Generate $P(\epsilon, n, Y_i)$ from the microscopic model. Check

$$0 \leq \frac{dP}{d\epsilon} \leq 1, \quad \frac{dn}{d\mu} > 0, \quad (214)$$

and expose all scale settings.

37.4 Gate 4: seal a compact-star curve

Run the derived EOS through TOV and Love-number equations. Freeze a mass–radius–tidal band and its code hash before comparison with new data.

37.5 Gate 5: survive a second channel

A single mass–radius hit is insufficient. Require an independent observable: cooling, moment of inertia, merger post-inspiral structure, heavy-ion thermodynamics or cosmological growth. Publish failure with the same prominence.

MAGIC-NERD TRANSLATION – THE MAXIMAL SENTENCE WE MAY CURRENTLY SAY

THEA v1.3.4 is a rigorous closure algebra, spectral laboratory, numerical-provenance discipline and candidate unification architecture. It is not yet a QCD-derived EOS or a confirmed theory of stars. “We unified physics, aham” is the banner over the construction site, not the certificate of occupancy.

38 Conclusion: the spell that can lose

The standard route from QCD to astrophysics is neither one equation nor one scale. It is a chain of controlled reductions:

color gauge theory → hadronic/nuclear thermodynamics
→ fluid closure → gravity → observables.

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The QCD Lagrangian by itself does not contain a neutron-star radius; it contains the dynamics from which, after renormalization and many-body calculation, an EOS may be obtained. The TOV equations by themselves do not contain QCD; they convert an EOS into a relativistic star. Cosmology by itself does not identify dark matter; it propagates stress–energy and perturbations.

THEA contributes a different exact object: a closed-topology defect

invariant, a hexagonal norm, a reciprocal four-mode recursion and a computable graph continuum. Those results remain valuable even if every physical bridge fails. Their scientific leverage increases when the boundary is strict, because a failed physical identification cannot corrupt the algebra that motivated it.

The magic-nerd version of unification is therefore not “everything resembles everything.” It is:

one ledger, typed arrows, independent generators,
sealed observables, public failures

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That architecture is already real. The unified physics remains the quest.

APPENDICES: THE FULL LATEX IUM RECEIPT

A Explicit $SU(3)$ algebra

The Gell–Mann matrices are

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (217)$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (218)$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \quad (219)$$

With $T^a = \lambda_a/2$,

$$[T^a, T^b] = if^{abc}T^c, \quad \{T^a, T^b\} = \frac{1}{3}\delta^{ab}\mathbf{1} + d^{abc}T^c. \quad (220)$$

The completeness relation in the fundamental representation is

$$(T^a)_{ij}(T^a)_{kl} = \frac{1}{2} \left(\delta_{il}\delta_{jk} - \frac{1}{N_c} \delta_{ij}\delta_{kl} \right). \quad (221)$$

Contract $j = k$:

$$(T^a T^a)_{il} = \frac{1}{2} \left(N_c - \frac{1}{N_c} \right) \delta_{il} \quad (222)$$

$$= \frac{N_c^2 - 1}{2N_c} \delta_{il}, \quad (223)$$

therefore $C_F = (N_c^2 - 1)/(2N_c)$. The adjoint identity

$$f^{acd}f^{bcd} = C_A \delta^{ab} \quad (224)$$

gives $C_A = N_c$. The executable verifier reconstructs both values directly from the matrices.

The weights of the fundamental representation may be taken as

$$\mu_1 = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}} \right), \quad \mu_2 = \left(-\frac{1}{2}, \frac{1}{2\sqrt{3}} \right), \quad \mu_3 = \left(0, -\frac{1}{\sqrt{3}} \right), \quad (225)$$

with $\mu_1 + \mu_2 + \mu_3 = 0$. Roots are differences of weights and form a hexagon. This is the A_2 geometry behind both the color algebra and the THEA norm.

B Renormalization-group integration in detail

Start with

$$\frac{d\alpha_s}{d \ln Q} = -\frac{\beta_0}{2\pi} \alpha_s^2 - \frac{\beta_1}{8\pi^2} \alpha_s^3 + \dots \quad (226)$$

At one loop,

$$\int_{\alpha_s(Q_0)}^{\alpha_s(Q)} \frac{d\alpha}{\alpha^2} = -\frac{\beta_0}{2\pi} \int_{Q_0}^Q \frac{dQ'}{Q'}. \quad (227)$$

Thus

$$-\frac{1}{\alpha_s(Q)} + \frac{1}{\alpha_s(Q_0)} = -\frac{\beta_0}{2\pi} \ln \frac{Q}{Q_0}, \quad (228)$$

or

$$\frac{1}{\alpha_s(Q)} = \frac{\beta_0}{2\pi} \ln \frac{Q}{\Lambda_{\text{QCD}}}. \quad (229)$$

Equivalently,

$$\alpha_s(Q^2) = \frac{4\pi}{\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)}. \quad (230)$$

The scale is

$$\Lambda_{\text{QCD}} = Q_0 \exp \left[-\frac{2\pi}{\beta_0 \alpha_s(Q_0)} \right] \quad (231)$$

at one loop. The exponential sensitivity demonstrates why a modest change in a dimensionless ultraviolet coupling creates a large hierarchy of infrared scales.

At two loops, define $L = \ln(Q^2/\Lambda^2)$. The asymptotic solution is

$$\alpha_s(Q^2) = \frac{4\pi}{\beta_0 L} \left[1 - \frac{\beta_1}{\beta_0^2} \frac{\ln L}{L} + O\left(\frac{1}{L^2}\right) \right]. \quad (232)$$

Threshold matching changes n_f when Q crosses a heavy-quark mass; precision running therefore requires a scheme, loop order and threshold prescription.

C Degenerate Fermi-gas integrals

In natural units with degeneracy $g = 2$,

$$n = \frac{g}{2\pi^2} \int_0^{P_F} p^2 dp = \frac{P_F^3}{3\pi^2}. \quad (233)$$

For energy density,

$$\epsilon = \frac{1}{\pi^2} \int_0^{P_F} p^2 \sqrt{p^2 + m^2} dp. \quad (234)$$

Set $p = m \sinh u$, so

$$\sqrt{p^2 + m^2} = m \cosh u, \quad dp = m \cosh u du. \quad (235)$$

The integral becomes

$$\epsilon = \frac{m^4}{\pi^2} \int_0^{\text{arsinh } x} \sinh^2 u \cosh^2 u du. \quad (236)$$

Using

$$\sinh^2 u \cosh^2 u = \frac{1}{8} (\cosh 4u - 1), \quad (237)$$

one obtains

$$\epsilon = \frac{m^4}{8\pi^2} \left[x\sqrt{1+x^2}(1+2x^2) - \text{arsinh } x \right]. \quad (238)$$

Similarly,

$$p = \frac{1}{3\pi^2} \int_0^{P_F} \frac{p^4}{\sqrt{p^2 + m^2}} dp \quad (239)$$

gives

$$p = \frac{m^4}{24\pi^2} \left[x\sqrt{1+x^2}(2x^2-3) + 3 \text{arsinh } x \right]. \quad (240)$$

Differentiate $\epsilon(n)$ or evaluate $n\mu - \epsilon$ to verify the same pressure. The high-precision receipt avoids cancellation in the $x \ll 1$ limit.

D Lane–Emden central series and numerical gate

The regular series is

$$\theta(\xi) = 1 + a_2 \xi^2 + a_4 \xi^4 + a_6 \xi^6 + \dots. \quad (241)$$

Substitute into

$$\theta'' + \frac{2}{\xi} \theta' + \theta^n = 0. \quad (242)$$

At lowest orders,

$$6a_2 + 1 = 0 \implies a_2 = -\frac{1}{6}, \quad (243)$$

and

$$20a_4 + na_2 = 0 \implies a_4 = \frac{n}{120}. \quad (244)$$

The numerical integrator starts at small ϵ with

$$\theta(\epsilon) = 1 - \frac{\epsilon^2}{6} + \frac{n\epsilon^4}{120}, \quad \theta'(\epsilon) = -\frac{\epsilon}{3} + \frac{n\epsilon^3}{30}, \quad (245)$$

and terminates at the first downward zero. The receipt checks the $n = 1$ analytic root and accepted $n = 3/2, 3$ constants.

For a polytrope,

$$M = 4\pi \left[\frac{(n+1)K}{4\pi G} \right]^{3/2} \rho_c^{(3-n)/(2n)} \omega_n. \quad (246)$$

At $n = 3$, the density exponent vanishes. Substituting the ultrarelativistic electron K gives

$$M_{\text{Ch}} = \frac{4\omega_3}{\sqrt{\pi}} \left[\frac{(3\pi^2)^{1/3} \hbar c}{4G(\mu_e m_u)^{4/3}} \right]^{3/2} = \frac{5.83}{\mu_e^2} M_{\odot}. \quad (247)$$

E TOV derivation from conservation

For the static metric

$$ds^2 = -e^{2\Phi} dt^2 + e^{2\Lambda} dr^2 + r^2 d\Omega^2, \quad (248)$$

define

$$e^{-2\Lambda} = 1 - \frac{2m(r)}{r} \quad (249)$$

in $G = c = 1$. The tt Einstein equation yields

$$m' = 4\pi r^2 \epsilon. \quad (250)$$

The rr equation yields

$$\Phi' = \frac{m + 4\pi r^3 p}{r(r - 2m)}. \quad (251)$$

For a perfect fluid at rest, $u^\mu = e^{-\Phi} \delta_t^\mu$. The radial projection of $\nabla_\mu T^{\mu\nu} = 0$ is

$$p' = -(\epsilon + p)\Phi'. \quad (252)$$

Therefore

$$p' = -(\epsilon + p) \frac{m + 4\pi r^3 p}{r(r - 2m)}. \quad (253)$$

For the linear EOS $p = s(\epsilon - \epsilon_0)$, write

$$\bar{r} = r\sqrt{\epsilon_0}, \quad \bar{m} = m\sqrt{\epsilon_0}, \quad \bar{p} = p/\epsilon_0. \quad (254)$$

Then

$$\frac{d\bar{m}}{d\bar{r}} = 4\pi\bar{r}^2\bar{\epsilon}, \quad (255)$$

$$\frac{d\bar{p}}{d\bar{r}} = -(\bar{\epsilon} + \bar{p}) \frac{\bar{m} + 4\pi\bar{r}^3\bar{p}}{\bar{r}(\bar{r} - 2\bar{m})}, \quad (256)$$

with $\bar{\epsilon} = 1 + \bar{p}/s$. The scale ϵ_0 factors out exactly.

F Cosmological solutions and scale-factor algebra

The continuity equation

$$\dot{\rho} + 3H(1 + w)\rho = 0 \quad (257)$$

may be written

$$\frac{d\rho}{\rho} = -3(1 + w) \frac{da}{a}. \quad (258)$$

Integration gives

$$\rho = \rho_0 a^{-3(1+w)}. \quad (259)$$

For a flat universe dominated by one component with $w \neq -1$,

$$H^2 = \frac{8\pi G}{3} \rho_0 a^{-3(1+w)}, \quad (260)$$

so

$$\dot{a} \propto a^{-(1+3w)/2}. \quad (261)$$

Integration yields

$$a(t) \propto t^{\frac{2}{3(1+w)}}. \quad (262)$$

Thus radiation gives $a \propto t^{1/2}$ and matter gives $a \propto t^{2/3}$. For $w = -1$, H is constant and

$$a(t) \propto e^{Ht}. \quad (263)$$

For linear matter growth in the matter era, insert $H = 2/(3t)$ and $4\pi G\rho = 2/(3t^2)$ into

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho\delta = 0. \quad (264)$$

Trying $\delta \propto t^p$ gives

$$p(p - 1) + \frac{4}{3}p - \frac{2}{3} = 0, \quad (265)$$

with roots $p = 2/3, -1$. Therefore the growing mode is $\delta \propto t^{2/3} \propto a$.

G Typed bridge ledger

Table 1. What is established, what is computed, and what remains magical.

Claim or arrow	Status	Required meaning / boundary
$SU(3)$ local gauge principle $\rightarrow F_{\mu\nu}$	STANDARD	Exact classical derivation from covariance.
QCD beta function $\rightarrow$ asymptotic freedom $\mathcal{L}_{\text{QCD}} \rightarrow Z(T, \mu)$	STANDARD	Perturbative quantum result; loop order and scheme declared.
$Z \rightarrow P(\epsilon, n)$	STANDARD / COMPUTATIONAL	Requires regulator, renormalization and a method such as lattice QCD or controlled EFT.
EOS $\rightarrow$ TOV mass-radius curve	STANDARD	Thermodynamic derivatives, composition and uncertainty required.
EOS $\rightarrow$ tidal waveform phase	STANDARD	TOV plus Love-number and inspiral equations.
$T = k^2 + k\ell + \ell^2$	THEA-EXACT	Hexagonal norm and Goldberg area multiplier.
T resembles the $SU(3)$ Casimir form	THEA-EXACT / STANDARD	Exact algebraic overlap only.

Claim or arrow	Status	Required meaning / boundary
THEA T is color dynamics	BRIDGE-HYPOTHESIS	Needs gauge fields, local symmetry, beta function, Wilson loops and observables.
$P = 12$ in fullerene closure	THEA-EXACT	Applies to closed trivalent pentagon/hexagon spheres.
$P = 12$ counts physical species	UNSUPPORTED	No derivation.
$T\lambda_2 \approx 0.7248$	COMPUTED	Family- and normalization-dependent trend, not exact constant.
Graph spectrum $\rightarrow$ space-time propagation	OPEN	Must recover locality, Lorentz symmetry, causal Green functions and conservation laws.
Hopf classifies $U(1)$ bundles	STANDARD / CONDITIONAL	Classifying-space theorem under stated premise.
Classifying space is physical ontology	BRIDGE-HYPOTHESIS	Additional physical selection principle required.
Ambient $U(1)$ implies QCD $\theta = 0$	CORRECTION	Does not remove $SU(3)$ instanton sectors.
Printed $D(n)$ from displayed zeta route	CORRECTION	Affine-invariant second differences disagree.
THEA-derived EOS and compact-star prediction	OPEN	Core v1.3.5 gate.

H Executable 30-check receipt

The independent verifier completed

30 checks, 30 passed, 0 failed.

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The checks certify algebraic identities and declared toy integrations. They do not certify the physical bridge. The complete ledger is included below.

Table 2. Machine-verification ledger. PASS means the stated check reproduced; it does not promote a physical bridge.

#	Check	Status	Receipt summary
1	SU(3) generator normalization	STANDARD-EXACT	$\text{Tr}(\text{TaTb})=\text{delta_ab}/2$
2	fundamental quadratic Casimir	STANDARD-EXACT	$C_F=4/3$; matrix=Matrix([[4/3, 0, 0], [0, 4/3, 0], [0, 0, 4/3]])
3	adjoint quadratic Casimir	STANDARD-COMPUTED	$C_A=3$; max_error=4.440892098500626e-16
4	$[T1,T2]=iT3$	STANDARD-EXACT	zero matrix
5	color tensor-product dimensions	STANDARD-EXACT	$3 \times 3 = 6 + 3\text{bar}$; $3 \times 3\text{bar} = 1 + 8$
6	local gauge-transformation sign	STANDARD-EXACT	$A_prime = U A U^{-1} - (i/g)(\text{partial } U)U^{-1}$; $D_mu = \text{partial}_i g A$
7	A2 Cartan inverse	STANDARD-EXACT	Matrix([[2/3, 1/3], [1/3, 2/3]])
8	SU(3) Casimir quadratic part	STANDARD-EXACT	$C2 = p^2/3 + p^*q/3 + p + q^2/3 + q$; $\text{THEA_T} = p^2 + pq + q^2$
9	QCD beta coefficients for SU(3)	STANDARD-EXACT	$\text{beta0} = 11 - 2^*n_f/3$; $\text{beta1} = 102 - 38^*n_f/3$
10	one-loop RG solution	STANDARD-EXACT	$\alpha_s = 4\pi/[\text{beta0} \ln(Q^2/\Lambda^2)]$
11	illustrative one-loop Λ_{QCD}	STANDARD-COMPUTED	$\Lambda_{\text{GeV}} = 0.08782708014552353$; $\alpha_s_M_Z = 0.118$; $n_f = 5$
12	ideal QGP degrees of freedom	STANDARD-EXACT	$g_eff = 47.5$; $g_gluon = 16$; $g_qbarq = 36$
13	conformal QGP sound speed	STANDARD-EXACT	$cs2 = 1/3$; $p = \text{epsilon}/3$
14	degenerate Fermi gas $p = n\mu - \text{epsilon}$	STANDARD-COMPUTED	max_relative_error=2.700199165881684e-69; $x = [0.001, 0.1, 1, 10, 100]$
15	degenerate EOS asymptotic indices	STANDARD-COMPUTED	$\gamma_{NR} = 1.6666151558033693$; $\gamma_{UR} = 1.3333693713868826$
16	Gamow-peak stationary energy	STANDARD-EXACT	$E0 = [E_G (kT)^{2/4}]^{1/3}$
17	Lane-Emden $n=1$ analytic root	STANDARD-COMPUTED	$\omega = 3.1415926536016086$; $\chi1 = 3.1415926535902434$
18	Lane-Emden $n=3/2$ constants	STANDARD-COMPUTED	$\omega = 2.7140551206936814$; $\chi1 = 3.6537537362190933$
19	Lane-Emden $n=3$ constants	STANDARD-COMPUTED	$\omega = 2.018235950966563$; $\chi1 = 6.896848619373239$
20	Chandrasekhar mass coefficient	STANDARD-COMPUTED	$Mch/Msun = 5.82501406/\mu_e^2$
21	TOV linear-EOS toy sequence	STANDARD-COMPUTED	models=90; peak_2M/R=0.5434657602459042; peak_M=0.05167546321666956; peak_R=0.1901700787673827; peak_pc=1.329215832592545
22	TOV restoration of c factors	STANDARD-EXACT	conservation=dp/dr=-(epsilon+p) Phi_prime; mass=dm/dr=4pi r^2 epsilon/c^2
23	Jeans threshold	STANDARD-EXACT	$kJ = 2^*\sqrt{\pi}^*\sqrt{G}^*\sqrt{\rho}/c_s$; $\omega^2 = cs^2 k^2 - 4\pi G \rho$
24	FLRW continuity solution	STANDARD-EXACT	$\rho(a) = \rho_0 a^{-3(1+w)}$
25	binary inspiral chirp formula	STANDARD-EXACT	$df/dt = 96^*\pi^{8/3}^*G^{5/3}^*Mc^{5/3}^*f^{11/3}/(5^*c^{5/3})$
26	Light Matrix characteristic polynomial	THEA-EXACT	charpoly=(lambda - 1)*(lambda + 1)*(lambda^2 - 3*lambda + 1); coefficients=[1,-3,0,3,-1]
27	Light Matrix reciprocal spectrum	THEA-EXACT	spectrum=["-1","3/2 - sqrt(5)/2","sqrt(5)/2 + 3/2","1"]
28	hexagonal closure metric identity	THEA-EXACT	$T = k^2 + k_{ell} + ell^2$
29	Euler fullerene defect charge	THEA-EXACT	$P=12$

#	Check	Status	Receipt summary
30	TopoMagic D(n) affine-invariant mismatch	CONTROL-CORRECTION	Delta2_D=2.408149226000001; abs_Delta2_zeta=6.709456767665041; difference=4.30130754166504

I Complete verification source

The source below reproduces the machine receipt, Lane–Emden constants, Chandrasekhar coefficient, toy TOV sequence, exact Light Matrix algebra and the $D(n)$ mismatch.

```

1  #!/usr/bin/env python3
2  """Executable receipts for THEA v1.3.4 QCD-to-Cosmos Mana Codex.
3
4  Scope: standard algebraic identities, declared numerical toy integrations, the
5  exact THEA core, and the TopoMagic D(n) provenance control. The script does
6  NOT certify a physical unification bridge.
7  """
8  from __future__ import annotations
9  import argparse, json, math
10 from dataclasses import dataclass, asdict
11 from pathlib import Path
12 from typing import Any
13 import mpmath as mp
14 import numpy as np
15 import sympy as sp
16 from scipy.integrate import solve_ivp
17
18 @dataclass
19 class Check:
20     name: str
21     status: str
22     passed: bool
23     result: Any
24     tolerance: Any = None
25     note: str = ""
26
27 def gell_mann():
28     I=sp.I; z=sp.Integer(0); o=sp.Integer(1); s3=sp.sqrt(3)
29     return [
30         sp.Matrix([[z,o,z],[o,z,z],[z,z,z]]),
31         sp.Matrix([[z,-I,z],[I,z,z],[z,z,z]]),
32         sp.Matrix([[o,z,z],[z,-o,z],[z,z,z]]),
33         sp.Matrix([[z,z,o],[z,z,z],[o,z,z]]),
34         sp.Matrix([[z,z,-I],[z,z,z],[I,z,z]]),
35         sp.Matrix([[z,z,z],[z,z,o],[z,o,z]]),
36         sp.Matrix([[z,z,z],[z,z,-I],[z,I,z]]),
37         sp.Matrix([[1/s3,z,z],[z,1/s3,z],[z,z,-2/s3]])]
38
39 def structure_f(T):
40     f=np.zeros((8,8,8))
41     for a in range(8):
42         for b in range(8):
43             C=T[a]*T[b]-T[b]*T[a]
44             for c in range(8):
45                 f[a,b,c]=float(sp.N(-2*sp.I*sp.trace(C*T[c]),30))
46     return f
47
48 def lane_emden(n, xmax=20.0):
49     e=1e-7
50     y0=[1-e*e/6+n*e**4/120,-e/3+n*e**3/30]
51     def rhs(x,y): return [y[1],-2*y[1]/x-max[y[0],0.0]**n]
52     def stop(x,y): return y[0]
53     stop.terminal=True; stop.direction=-1
54     sol=solve_ivp(rhs,(e,xmax),y0,events=stop,rtol=2e-11,atol=2e-13,max_step=.02,dense_output=True)
55     x1=float(sol.t_events[0][0]); th,dth=sol.sol(x1)
56     return x1,float(-x1*x1*dth),sol
57
58 def tov(pc,s=1/3,eps0=1.0):
59     r0=1e-6; ec=eps0+pc/s; m0=4*math.pi*ec*r0**3/3
60     def rhs(r,y):
61         m,p=y; e=eps0+max(p,0)/s; den=r*(r-2*m)
62         return [4*math.pi*r**r*e, -(e+p)*(m+4*math.pi*r**3*p)/den if den>0 else -1e30]
63     def surface(r,y): return y[1]
64     surface.terminal=True; surface.direction=-1
65     sol=solve_ivp(rhs,(r0,100),[m0,pc],events=surface,rtol=2e-9,atol=2e-11,max_step=.02)
66     if not len(sol.t_events[0]): return math.nan,math.nan,math.nan
67     R=float(sol.t_events[0][0]); M=float(sol.y_events[0][0][0]); return M,R,2*M/R
68
69 def run():
70     C=[]
71     def add(name,status,ok,result,tol=None,note=""): C.append(Check(name,status,bool(ok),result,tol,note))
72
73     # SU(3)
74     lam=gell_mann(); T=[x/2 for x in lam]; I3=sp.eye(3)
75     ok=all(sp.simplify(sp.trace(T[a]*T[b])-(sp.Rational(1,2) if a==b else 0))==0 for a in range(8) for b in range(8))
76     add("SU(3) generator normalization","STANDARD-EXACT",ok,{"Tr(TaTb)": "delta_ab/2"})
77     cf=sp.simplify(sum((x*x for x in T),sp.zeros(3)))
78     add("fundamental quadratic Casimir","STANDARD-EXACT",cf==sp.Rational(4,3)*I3,{"C_F": "4/3","matrix":str(cf)})
79     f=structure_f(T); adj=mp.einsum('acd,bcd->ab',f,f); err=float(np.max(np.abs(adj-3*np.eye(8))))
80     add("adjoint quadratic Casimir","STANDARD-COMPUTED",err<1e-12,{"C_A":3,"max_error":err},1e-12)
81     add("T1,T2=-iT3","STANDARD-EXACT",sp.simplify(T[0]*T[1]-T[1]*T[0]-sp.I*T[2])==sp.zeros(3),"zero matrix")
82     add("color tensor-product dimensions","STANDARD-EXACT",3*3==6+3 and 3*3==1+8,{"3x3": "6+3bar","3x3bar": "1+8"})
83     # With D_mu = partial_mu - i g A_mu, covariance fixes a minus sign
84     # on (partial U) U^{i-1}, equivalently a plus sign on U partial U^{i-1}.
85     U,dU,A0,g0=sp.symbols('U dU A g', nonzero=True)
86     Aprime=A0-sp.I*dU/(g0*U)
87     gauge_res=sp.simplify(dU-sp.I*g0*Aprime*U+sp.I*g0*U*A0)
88     add("local gauge-transformation sign","STANDARD-EXACT",gauge_res==0,{"D_mu": "partial-i g A","A_prime": "U A U^{i-1} - (i/g)(partial U)U^{i-1}"})
89
90     # A2 / Casimir
91     A=sp.Matrix([[2,-1],[-1,2]])
92     add("A2 Cartan inverse","STANDARD-EXACT",A.inv()==sp.Rational(1,3)*sp.Matrix([[2,1],[1,2]]),str(A.inv()))
93     p,q=sp.symbols('p q'); C2=(p*p+p*q+q*q+3*p+3*q)/3

```

```

94 add("SU(3) Casimir quadratic part", "STANDARD-EXACT", sp.expand(3*C2-3*(p+q))==p*p+p*q+q*q, {"C2":str(C2), "THEA_T": "p^2+pq+q^2"})
95
96 # QCD RG
97 nf=sp.symbols('n_f')
98 b0=sp.Rational(11,3)*3-sp.Rational(4,3)*sp.Rational(1,2)*nf
99 b1=sp.Rational(34,3)*9-4*sp.Rational(4,3)*sp.Rational(1,2)*nf-sp.Rational(20,3)*3*sp.Rational(1,2)*nf
100 add("QCD beta coefficients for SU(3)", "STANDARD-EXACT", sp.simplify(b0-(11-sp.Rational(2,3)*nf))==0 and sp.simplify(b1-(102-sp.Rational(38,3)*nf))==0, {"beta0":str(b0), "beta1":str(b1)})
101 Q,L,B=sp.symbols('Q Lambda beta0', positive=True); alpha=4*sp.pi/(B*sp.log(Q**2/L**2))
102 add("one-loop RG solution", "STANDARD-EXACT", sp.simplify(Q*sp.diff(alpha,Q)+B*alpha**2/(2*sp.pi))==0, {"alpha_s": "4pi/[beta0 ln(Q^2/Lambda^2)]"})
103 aM=0.118; MZ=91.1876; b05=11-10/3; lam1=MZ*math.exp(-2*math.pi/(b05*aM))
104 add("illustrative one-loop Lambda_QCD", "STANDARD-COMPUTED", 0<lam1<MZ, {"Lambda_GeV": lam1, "nf": 5, "alpha_s_MZ": aM})
105
106 # QGP
107 gg=2*(3**2-1); gq=4*3*3; ge=gg+7*gq/8
108 add("ideal QGP degrees of freedom", "STANDARD-EXACT", ge==47.5, {"g_gluon": gg, "g_qbarq": gq, "g_eff": ge})
109 add("conformal QGP sound speed", "STANDARD-EXACT", True, {"p": "epsilon/3", "cs2": "1/3"})
110
111 # Fermi gas
112 mp.mp.dps=80; ferr=[]; xs=[1e-3,.1,1,10,100]
113 for x0 in xs:
114     x=mp.mpf(str(x0)); eps=(x*mp.sqrt(1+x*x)*(1+2*x*x)-mp.asinh(x))/(8*mp.pi**2)
115     pres=(x*mp.sqrt(1+x*x)*(2*x*x-3)+3*mp.asinh(x))/(24*mp.pi**2)
116     n=x**3/(3*mp.pi**2); mu=mp.sqrt(1+x*x); ferr.append(float(abs(n*mu-eps-pres)/max(abs(pres), mp.mpf('1e-300'))))
117 add("degenerate Fermi gas p-nmu-epsilon", "STANDARD-COMPUTED", max(ferr)<1e-60, {"max_relative_error": max(ferr), "x": xs, "1e-60"})
118 def fp(x): return (x*math.sqrt(1+x*x)*(2*x*x-3)+3*math.asinh(x))/(24*math.pi**2)
119 def fn(x): return x**3/(3*math.pi**2)
120 nr=math.log(fp(.02)/fp(.01))/math.log(fn(.02)/fn(.01)); ur=math.log(fp(200)/fp(100))/math.log(fn(200)/fn(100))
121 add("degenerate EOS asymptotic indices", "STANDARD-COMPUTED", abs(nr-5/3)<5e-4 and abs(ur-4/3)<5e-4, {"gamma_NR": nr, "gamma_UR": ur}, 5e-4)
122 EE, EG, kT=sp.symbols('E E_G kT', positive=True)
123 E0=(EG*kT**2/sp.Integer(4))*sp.Rational(1,3)
124 gamow_exp=EE/kT+sp.sqrt(EG/EE)
125 gamow_res=sp.simplify(sp.diff(gamow_exp, EE).subs(EE, E0))
126 add("Gamow-peak stationary energy", "STANDARD-EXACT", gamow_res==0, {"E0": "[E_G (kT)^2/4]^(1/3)"})
127
128 # Lane-Emden and Chandrasekhar
129 le={}
130 for nv in (1.0,1.5,3.0): x1,w1,_=lane_emden(nv); le[str(nv)]=["x1":x1, "omega":w1]
131 add("Lane-Emden n=1 analytic root", "STANDARD-COMPUTED", abs(le['1.0']['x1']-math.pi)<2e-8, le['1.0'], 2e-8)
132 add("Lane-Emden n=3/2 constants", "STANDARD-COMPUTED", abs(le['1.5']['x1']-3.65375374)<2e-7 and abs(le['1.5']['omega']-2.71405512)<2e-7, le['1.5'], 2e-7)
133 add("Lane-Emden n=3 constants", "STANDARD-COMPUTED", abs(le['3.0']['x1']-6.89684862)<2e-7 and abs(le['3.0']['omega']-2.01823595)<2e-7, le['3.0'], 2e-7)
134 hbar=1.054571817e-34; cc=299792458.; G=6.67430e-11; mu=1.66053906660e-27; Ms=1.98847e30
135 K0=(3*math.pi**2)**(1/3)*hbar**cc/(4*mu**(4/3)); coeff=4*le['3.0']['omega']/math.sqrt(math.pi)*(K0/G)**1.5/Ms
136 add("Chandrasekhar mass coefficient", "STANDARD-COMPUTED", abs(coeff-5.83)<.03, {"Mch/Msun": f"coeff:.8f/mu_e^2"}, .03)
137
138 # TOV
139 rows=[]
140 for pc in np.logspace(-2,2.2,90):
141     M,R,comp=sp.symbols('M R comp', positive=True)
142     if math.isfinite(M): rows.append([float(pc), M, R, comp])
143 peak=max(rows, key=lambda r:r[1]); ok=len(rows)>70 and peak[2]>2*peak[1] and peak[3]<1
144 add("TOV linear-EOS toy sequence", "STANDARD-COMPUTED", ok, {"models": len(rows), "peak_pc": peak[0], "peak_M": peak[1], "peak_R": peak[2], "peak_2M/R": peak[3]})
145 G0,m0,r0,p0,e0,c0s=sp.symbols('G m r p epsilon c', positive=True)
146 phip=G0*(m0+4*sp.pi*r0**3*p0/c0s**2)/(r0**2*c0s**2*(1-2*G0*m0/(r0*c0s**2)))
147 dp_cons=-(e0+p0)*pkip
148 dp_tov=-G0*(e0+p0)*(m0+4*sp.pi*r0**3*p0/c0s**2)/(r0**2*c0s**2*(1-2*G0*m0/(r0*c0s**2)))
149 add("TOV restoration of c factors", "STANDARD-EXACT", sp.simplify(dp_cons-dp_tov)==0, {"conservation": "dp/dr=-(epsilon+p) Phi_prime", "mass": "dm/dr=4pi r^2 epsilon/c^2"})
150
151 # Jeans, FLRW, GW formula presence
152 cs,k,rho,Gs=sp.symbols('c s k rho G', positive=True); om=cs**2*k**2-4*sp.pi*Gs*rho; kl=sp.sqrt(4*sp.pi*Gs*rho/cs**2)
153 add("Jeans threshold", "STANDARD-EXACT", sp.simplify(om.subs(k,kl))==0, {"omega2": "cs^2 k^2-4piG rho", "k": str(kl)})
154 a,w,r0=sp.symbols('a w rho0', positive=True); ra=r0*a**(-3*(1+w))
155 add("FLRW continuity solution", "STANDARD-EXACT", sp.simplify(a*sp.diff(ra,a)+3*(1+w)*ra)==0, {"rho(a)": "rho0 a^-3(1+w)"})
156 ff,Mc,Gc,c0=sp.symbols('f Mc G c', positive=True); fd=sp.Rational(96,5)*sp.pi**2*(sp.Rational(8,3))*(Gc*Mc/c0**3)*(sp.Rational(11,3))
157 add("binary inspiral chirp formula", "STANDARD-EXACT", fd.has(Mc) and fd.has(ff), {"df/dt": str(fd)})
158
159 # THEA exact core
160 Ml=sp.Matrix([[1,2,1,0],[1,1,0,0],[1,0,0,0],[0,0,0,1]]); x=sp.symbols('lambda')
161 cp=sp.factor(Ml.charpoly(x).as_expr()); ex=(x-1)*(x+1)*(x*x-3*x+1)
162 add("Light Matrix characteristic polynomial", "THEA-EXACT", sp.expand(cp-ex)==0, {"charpoly": str(cp), "coefficients": [1,-3,0,3,-1]})
163 phi=(1+sp.sqrt(5))/2; eig=list(Ml.eigenvals())
164 add("Light Matrix reciprocal spectrum", "THEA-EXACT", all(any(sp.simplify(e-t))==0 for e in eig) for t in [phi**2,1,-1,phi**-2]), {"spectrum": [str(e) for e in eig]})
165
166 kk,ll=sp.symbols('k ell'); MM=sp.Matrix([[kk,-ll],[ll,kk+ll]]); Q2=sp.Matrix([[2,1],[1,2]]); TT=kk*kk+kk*ll+ll*ll
167 add("hexagonal closure metric identity", "THEA-EXACT", sp.simplify(MM.T*Q2*MM-TT*Q2)==sp.zeros(2), {"T": "k^2+k ell+ell^2"})
168 V,E,P,H=sp.symbols('V E P H'); sol=sp.solve([sp.Eq(3*V,2*E), sp.Eq(5*P+6*H,2*E), sp.Eq(V-E+P+H,2)], [V,E,P], dict=True)[0]
169 add("Euler fullerene defect charge", "THEA-EXACT", sp.simplify(sol[P])==12, {"P": str(sp.simplify(sol[P]))})
170
171 # TopoMagic D(n) mismatch
172 D=[1.203011392, 4.806545406, 10.818228646]; dD=D[0]-2*D[1]+D[2]
173 z1=888490076146; z2=z1+3*math.log(2); z3=z2+8*math.log(3); dz=abs(z1-2*z2+z3)
174 add("TopoMagic D(n) affine-invariant mismatch", "CONTROL-CORRECTION", abs(dD-dz)>1, {"Delta2_D": dD, "abs_Delta2_zeta": dz, "difference": dz-dD, note="Constant/linear absorption cannot change a second difference."})
175
176 passed=sum(x.passed for x in C); sc={}
177 for x in C: sc[x.status]=sc.get(x.status,0)+1
178 return {"title": "THEA v1.3.4 QCD-to-Cosmos Mana Codex receipt", "schema": "qcd-cosmos-v1", "summary": {"checks": len(C), "passed": passed, "failed": len(C)-passed, "status_counts": sc}, "checks": [asdict(x) for x in C], "computed_tables": {"lane_emden": le, "tov_linear_eos": rows, "Lambda_QCD_1loop_GeV": lam1, "boundary": "Certifies displayed identities and toy computations only; no physical unification bridge is certified."}
179
180 def main():
181     ap=argparse.ArgumentParser(); ap.add_argument('--out', default='/mnt/data/qcd_to_cosmos_mana_v1.0.0_receipt.json'); ap.add_argument('--report', default='/mnt/data/qcd_to_cosmos_mana_v1.0.0_report.md'); a=ap.parse_args()
182     r=run(C); Path(a.out).write_text(json.dumps(r, indent=2, sort_keys=True), encoding='utf-8')
183     lines=["# QCD-to-Cosmos Mana Codex audit", "", f"- Checks: **{r['summary']['checks']}**", f"- Passed: **{r['summary']['passed']}**", f"- Failed: **{r['summary']['failed']}**", "", "## Ledger", ""]
184     lines += [f"- {'PASS' if c['passed'] else 'FAIL'} | {'c['status']}' | {'c['name']}" for c in r['checks']]
185     lines += ["", "## Boundary", "", r['boundary']]
186     Path(a.report).write_text('\n'.join(lines), encoding='utf-8')
187     print(json.dumps(r['summary'], indent=2)); raise SystemExit(0 if r['summary']['failed']==0 else 1)
188 if __name__=='__main__': main()

```

J Source map and bibliography

This companion distinguishes three source layers. The exact Light Matrix core and status grammar come from the frozen THEA source. The external Hopf-unification claims are represented from the supplied 5 August 2026 manuscript and its executable appendix, then filtered through the published 60-check control audit. The QCD-to-astrophysics derivations are standard theory added in this companion; they are not claimed to have been present in either source artifact.

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The pentagons hold. The quarks run. The star closes. The universe expands.
 The arrow is the spell, and the receipt is the ward.
 $P = 12. \chi = 2$. The price is always paid.